

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Refrigeration & Air Conditioning
Laboratory Manual
Course Code: ME344
B.Tech 6th Sem

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ME344 Refrigeration & Air Conditioning	
CO1	Understand the principles and applications of refrigeration systems
CO2	Evaluate performance of Vapour compression refrigeration system.
CO3	Apply working principle of VAR/VCR system to solve numericals based on VCR and VAR system.
CO4	Understand basics of psychrometry, air conditioning processes and different air conditioning systems.
CO5	Analyze different psychrometric processes on general cycle air conditioning trainer.

List of Experiments (ME 344 RAC)		
Sr. No.	Title	Course Outcomes
1	Study of Simple Vapor compression Refrigeration System and its components.	CO1
2	Performance test on General cycle refrigeration trainer.	CO1, CO5
3	Study of Cascade Refrigeration system for producing low temperature.	CO1, CO5
4	Study of Vapor Absorption refrigeration system.	CO4
5	Study of different psychrometric terms and processes.	CO5
6	Performance Test on General Cycle air-conditioning trainer.	CO5
7	Study of different air-conditioning systems.	CO7
8	Design of Air Conditioning System and load calculation for residential and commercial buildings.	CO3, CO6
9	Study and Performance on Sling psychrometer.	CO1
10	Exhaust gas heat recovery type VAR system.	CO2, CO6
	Study of measurement devices of all experimental set-ups used in RAC laboratory.	

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CERTIFICATE

This is to certify that Mr. /Ms. _____ of _____ Class,
Roll No. _____ Exam No. _____ has satisfactorily completed his / her term work in
_____ for the term ending _____ in 20____ /
20____.

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Sign of the Faculty Head of the Department

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 CHAMOS Matrusanstha Department of Mechanical Engineering
Subject: Refrigeration and Air Conditioning (ME 344) **Semester: 6th**

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Refrigeration and Air-Conditioning (ME307)

Date:

Experiment No 1
Simple Vapour Compression Refrigeration System

AIM: Study of simple vapor Compression Refrigeration System (VCRS) and its components.

OBJECTIVES:

1. To study various Components of Vapour Compression Refrigeration
2. System (VCRS) like Compressor, Condenser, Evaporator and Expansion device.
3. To study effect of sub cooling and superheating on the performance of VCRS.
4. To Study the effect of varying the suction and discharge pressure on the performance of VCRS.

THEORY:

Mainly there are four components in VCRS.

(1) Evaporator (2) Compressor (3) Condenser (4) Expansion Device.

First the refrigerant is compressed adiabatically in the compressor. The pressure and temperature of the refrigerant will increase. Then refrigerant will enter in the Condenser. Heat will be rejected from the refrigerant to the atmosphere in Condenser at constant pressure. Then, the refrigerant will enter in expansion device where expansion takes place which result in reduction of temperature and pressure of refrigerant. This low temperature refrigerant will enter into evaporator and by its evaporation cooling will be produced at constant pressure. The heat absorbed in evaporator is termed as Refrigeration effect. This low temperature refrigerant vapor enters in compressor and cycle is repeated.

QUESTIONS:

1. Define Refrigeration and Air-conditioning. State the application of refrigeration.
2. Explain with neat sketch the working principle of simple Vapor Compression Refrigeration System.
3. What is C.O.P of a refrigeration system?
4. Explain the effect of Sub-cooling and Super-heating of on the performance of a simple VCRS.
5. Explain the effect of varying suction pressure and discharge pressure on the performance of VCR with p-h and T-S diagram.
6. Draw and explain actual vapor compression refrigeration cycle.
7. The following results were obtained in a test conducted on vapor-compression refrigeration system. Evaporator temperature = 28°C, Condenser pressure = 2.8 bar
Refrigerant entering the condenser = 3°C superheat

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Refrigerant leaving the condenser = 12°C

Draw up heat balance sheet and calculate C.O.P. Take the following properties for refrigerant

Saturation Temp °C	Pressure bar	Enthalpy kJ/kg		Entropy kJ/kg K vapour	Sp. Heat kJ/kg K	
		liquid	vapour		liquid	vapour
14	2.8	438.3	803.5	5.533	1.382	0.67
-28	0.4	381.9	783.8	5.686	-	-

8. A R-12 vapour compression system has saturated suction temp of -5°C and saturated discharge temp of 40°C. calculate (1) mass flow rate /ton of refrigeration (2) theoretical piston displacement /ton of refrigeration (3) theoretical power /ton of refrigeration (4) COP with (a) dry compression (b) wet compression. Find the system performance when capacity is 15TR.
9. A food storage locker required a refrigeration system of a 2400 kJ/min capacity. The evaporator temp is -10°C and a condenser temp is +30°C. The refrigerant used is R12. The refrigerant is sub cooled by 6°C before entering the expansion valve. The vapour is superheated by 7°C before leaving the evaporator coil. The compression of the refrigerant is isentropic. The refrigerant compressor is 2 cylinder, single acting L/D=1.25 and it operates at 1000 rpm. The properties of R12 is as under:

Saturation Temp °C	Absolute Pressure bar	Sp. volume m³/kg v _g	Enthalpy kJ/kg		Entropy kJ/kg K		Sp. Heat kJ/kg K	
			liquid	vapour	liquid	vapour	liquid	vapour
+30	7.4379	0.02376	228.89	365.16	1.0992	1.5487	0.998	0.721
-10	2.1893	0.07731	190.78	348.17	0.9658	1.5639	0.911	0.611

CONCLUSION:

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Refrigeration and Air-Conditioning (ME307)

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Experiment No 2
Performance Test General Cycle Refrigeration Trainer

Refrigeration is the process of removing the heat from a substance, which is at ambient temperature and to lower its temperature upto desired level depending upon the application.

Based on the desired temperature, Refrigeration generally classified in three groups: high temperature, Medium temperature and low temperature group. High temperature group is one in which the object temperature is at 22°–26°C. e.g. air conditioning system. In medium temperature, object temperature ranges between 4°C to 20°C, e.g. milk chiller or bottle coolers. In low temperature group, temperatures are kept below 0°C to lower range of -20°C. applications of these are household refrigerators or ice making plants. Some applications also need the temperatures of the order of -100°C to produce such cold temperatures with simple refrigerator is not possible and it needs use of cascade refrigerating systems.

In refrigeration, two systems are used, namely vapour absorption system and vapour compression system. The difference between the systems lies in the way of rising the pressure of low pressure vapour. The absorption system dissolves refrigerant in solvent and raises its pressure by a pump while in compression system compressor compresses the refrigerant vapour.

The unit consists of vapour compression refrigeration cycle. The compressor sucks the low pressure vapour from evaporator, compresses it to high pressure and sends it to condenser. Temperature of vapour rises due to compression. The high pressure, high temperature vapour gets condensed in condenser, as it rejects heat to cooling medium flowing over the condenser tubes. The high pressure liquid issuing from condenser goes to expansion device, where it is throttled to low pressure. Due to throttling, its temperature decreases, this low pressure, low temperature wet vapour goes to evaporator where it boils and picks up its latent heat of evaporation from evaporator, thus cooling the objects kept there in. this vapour is again sucked by compressor, compressed and sent to condenser.

SPECIFICATIONS

COMPRESSOR : - Kirloskar make Hermetically Sealed Compressor, having capacity of 0.3 TR.

REFRIGERANT:-R-12 (Di- chloro di-fluoro methane- CCl_2F_2)

Finned tube air cooled condenser with fan.

EXPANSION DEVICES:- i) Thermostatic Expansion valve, ii) Capillary tube.

Evaporator coil with water circulation for balancing refrigerating effect.

CONTROLS:

Service valves

Solenoid valve

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Filter-drier for refrigerant
High/low pressure cutout
Thermostat

MEASUREMENTS:

Thermometers:- 10 °C to 110 °C – 6 nos
Pressure gauges for condensing and evaporating pressure.
Energy meter for compressor.
Voltmeter and ammeter for compressor.
Dial thermometer for temperature of water in evaporator.
Rota meter for liquid refrigerant flow measurement.
Measuring jug & stop watch.

PRINCIPAL OF OPERATION:

The apparatus consists of a medium temperature vapour compression refrigeration cycle. It moves that from the water surrounding the evaporator coil, and it lowers the temperature of water. If the water is continuously circulated over the evaporator coil, then temperature of water at outlet will remain constant. If some means is provided to measure temperature drop of water and water flow rate when water temperatures are steady, then heat given by water is equivalent to the refrigerating effect. The apparatus works on this principle. This is utilized in finding capacity of the system and actual C.O.P. of the system.

EXPERIMENTAL PROCEDURE:

1. Start water flow in the evaporator. Keep water flow about 3 to 4 lit/min.
2. Connect mains supply to the unit.
3. Depending upon expansion device to be used, open inlet valve for thermostatic expansion valve (or capillary) and close valve for the other. Switch on solenoid valve, if experiment is to be conducted on thermostatic expansion valve. Start condenser fan and compressor.
4. Wait till the temperatures and pressure becomes steady. It should take around 20 to 30 minutes.
5. Check time required for compressor energy meter 10 imps.
6. As the evaporator temperature remains steady, note down the readings and fills up observation table.
7. Repeat the experiment with the other expansion device.

OBSERVATIONS:

Expansion device used-Capillary/ thermostatic expansion valve.
Time for 10 imps of Compressor Energy meter, t_c (sec)
Evaporator water flow – time for 1 lit. (sec)
Supply voltage, volts.
Compressor current, amp.

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Temperatures

a) Refrigerant temperatures –

- (i) Condenser inlet, t_{ci} , °C.
- (ii) Condenser outlet, t_{co} , °C
- (iii) Evaporator inlet, t_{ei} , °C
- (iv) Evaporator outlet, t_{eo} , °C

b) Water temperatures-

- (i) evaporator inlet, t_{wi} °C
- (ii) evaporator outlet, t_{wo} °C

Condensing pressure P_c , kg/cm²

Evaporating pressure P_e , kg / cm²

Refrigerant flow Q , LPH.

CALCULATIONS:

1. REFRIGERATING EFFECT IS BALANCED BY WATER CIRCULATION:

so, Heat given by water = Refrigerating effect.

$$m \cdot C_p \cdot \Delta T = \text{Refrigerating effect} = \text{_____}$$

where m = mass flow rate of water = 1 / time for 1 lit. kg/s
 (density of water is assumed as 1 kg / m³)

C_p = Specific heat of water = 4.2 kJ / kg

$$\Delta T = t_{wi} - t_{wo}$$

2. COMPRESSOR WORK

$$C_w = \frac{3600n}{t \times C_c} (k) = \text{_____}$$

Where, t_c = Time for compressor energy meter disc, sec.

EMCC = Compressor energy meter constant = 3200 Rev/Kwh

3. ACTUAL C.O.P.

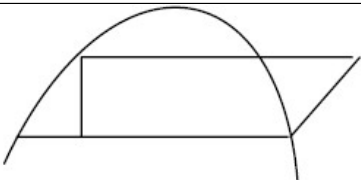
$$COP = \frac{R.E.}{W_w} = \text{_____}$$

4. THEORETICAL C.O.P.

Condensing Pressure = (Gauge reading x 0.981) + 1.014 bar abs.

Evaporating pressure = (Gauge reading x 0.981) + 1.014 bar abs.

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Plot the cycle on P-h chart as shown & find out- $H_{ei} =$ $H_{co} = H_{ei} =$ $H_{eo} =$ $R.E. \text{ theo} = H_{eo} - H_{ei}$ $CW \text{ theo} = H_{ei} - H_{eo}$ $C.O.P. \text{ theo} = H_{eo} - H_{ei} / H_{ei} - H_{eo} = \underline{\hspace{2cm}}$	
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5. CARNOT C.O.P.

For the cycle operating between the saturation temperatures of T_H and T_L , where T_H and T_L are saturation temperatures in K, corresponding to condensing and evaporating pressures respectively,

$$COP \text{ (carnot)} = T_L / (T_H - T_L) = \underline{\hspace{2cm}}$$

6. MASS FLOW RATE OF REFRIGERENT

Liquid refrigerant flow, $Q = \underline{\hspace{2cm}}$ LPH

From table of properties of R-12 find v , specific volume of liquid at condensing pressure

Mass flow rate –

$$m = Q / (1000 \times v) = \underline{\hspace{2cm}} \text{ kg / hr}$$

7. THEORETICAL REFRIGERATING EFFECT

$$RE \text{ Theo} = m (H_{eo} - H_{ei}) \text{ kJ/s (kW)} = \underline{\hspace{2cm}}$$

8. CAPACITY OF THE SYSTEM

$$P = R.E. / 3.5 \text{ Tons of refrigeration}$$

9. VOLUMETRIC EFFICIENCY OF COMPRESSOR

Compressor swept volume is $V_2 = 2.18 \text{ m}^3/\text{hr}$

Volume discharged by compressor is $V_1 = \underline{\hspace{2cm}} \text{ m}^3/\text{hr}$

$$V_1 = m \times V_s$$

Where, $m =$ mass flow rate = $\underline{\hspace{2cm}}$

$V_s =$ sp. Volume of vapour at suction (evaporation) pressure = $\underline{\hspace{2cm}}$

$$\eta = \frac{V_1}{V_2}$$

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THERMODYNAMIC PROPERTIES OF R-12

TEMP °C	SPECIFIC VOLUME m ³ / kg	
	Sat. liquid	Sat. vapour
-10	0.0007000	0.07665
-5	0.0007078	0.06496
0	0.0007159	0.05539
4	0.0007227	0.04895
8	0.0007297	0.04340
12	0.0007370	0.03860
16	0.0007446	0.03442
20	0.0007525	0.03078
24	0.0007607	0.02759
26	0.0007650	0.02614
28	0.0007694	0.02478
30	0.0007739	0.02352
32	0.0007785	0.02231
34	0.0007832	0.02118
36	0.0007880	0.02012
38	0.0007929	0.01912
40	0.0007980	0.01817

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OBSERVATION TABLE

SR. NO.	DESCRIPTION	CAPILLARY TUBE	THERMOSTATIC EXPANSION VALVE
1	Time for 10 rev. of		
2	Evap. Water flow 1 lit sec		
3	Supply voltage “V”		
4	Compressor current “Amp”		
5	Cond. Inlet tci °c		
6	Cond. Outlet tco °c		
7	Evap. Inlet tei °c		
8	Evap. Outlet teo °c		
9	Evap. Water Inlet twi °c		
10	Evap water Outlet two °c		
11	Cond pressure Pc kg/cm ²		
12	Evap. Pressure Pe kg /cm ²		
13	Ref. Flow Q LPH		

Conclusion:

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Refrigeration and Air-Conditioning (ME307)

Date:

Experiment No 3
Cascade Refrigeration

AIM: STUDY OF CASCADE REFRIGERATION SYSTEM FOR PRODUCING LOW TEMPERATURE.

OBJECTIVES:

1. To study Cascade refrigeration system.
2. Comparison of VCRS and Cascade Refrigeration System.
3. Advantages of Cascade Refrigeration System over Compound Compression System and VCRS.

THEORY:

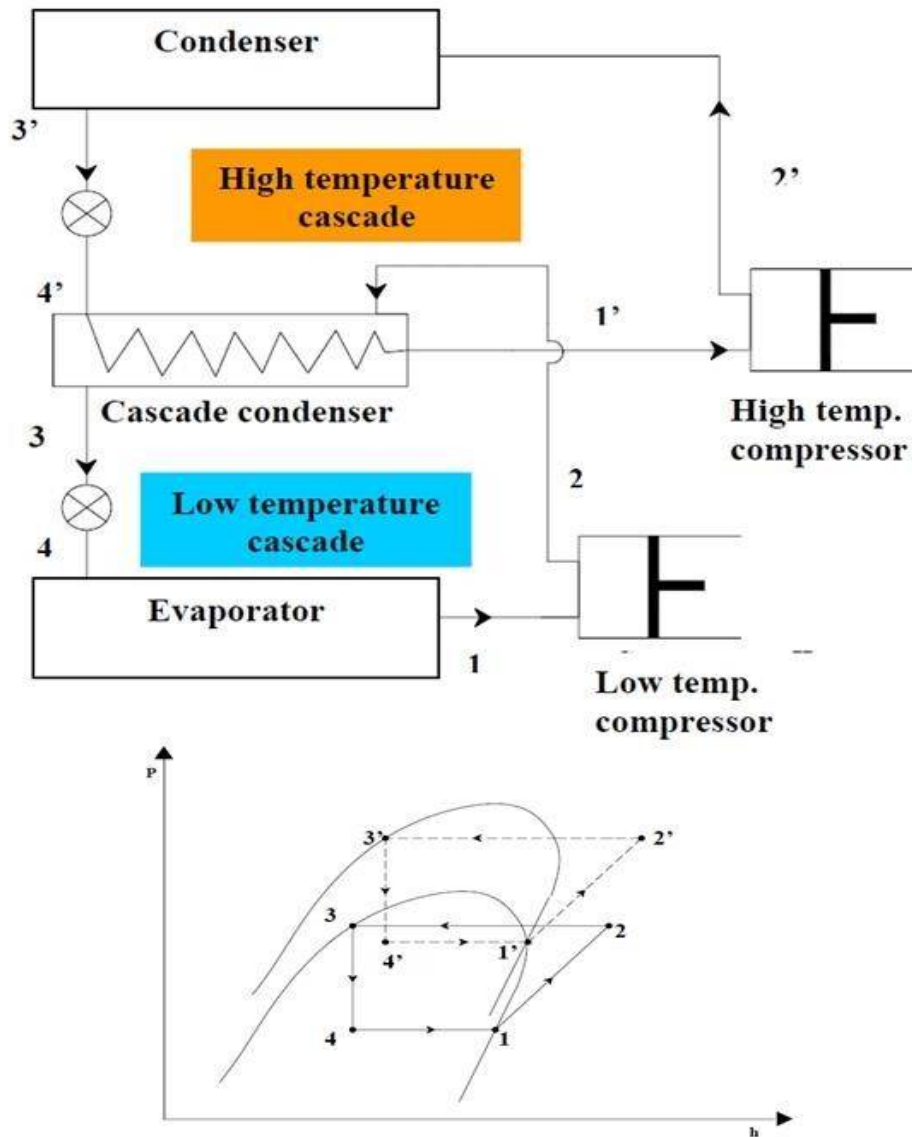
Cascade refrigeration system is the extension of VCRS for producing low temperature which is the main constrain in VCRS. In a cascade system a series of refrigerants with progressively lower boiling points are used in a series of single stage units. The condenser of lower stage system is coupled to the evaporator of the next higher stage system and so on. The component where heat of condensation of lower stage refrigerant is supplied for vaporization of next level refrigerant is called as cascade condenser. Figures show the schematic and P-h diagrams of a two-stage cascade refrigeration system. As shown, this system employs two different refrigerants operating in two individual cycles. They are thermally coupled in the cascade condenser. The refrigerants selected should have suitable pressure-temperature characteristics. An example of refrigerant combination is the use of carbon dioxide (NBP = -78.4°C , $T_{\text{cr}} = 31.06^{\circ}\text{C}$) in low temperature cascade and ammonia (NBP = -33.33°C , $T_{\text{cr}} = 132.25^{\circ}\text{C}$) in high temperature cascade. It is possible to use more than two cascade stages, and it is also possible to combine multi-stage systems with cascade systems.

APPLICATION OF THE CASCADE REFRIGERATION SYSTEM.

- a. Liquefaction of petroleum vapour
- b. Liquefaction of industrial gases
- c. Manufacturing of dry ice
- d. Deep freezing

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BLOCK DIAGRAM and P-h CHART OF CASCADE REFRIGERATION SYSTEM



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CONCLUSION:

QUESTIONS:

1. Comparison of cascade refrigeration system with VCRS and Compound Compression.
2. Explain 2 stage cascade refrigeration system by block diagram and p-h diagram.
3. Applications of cascade refrigeration system.

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Date:

Experiment No 4
Vapor Absorption Refrigeration System.

AIM: TO STUDY TWO TYPES OF VAPOR ABSORPTION REFRIGERATION SYSTEM.

- 1. $\text{NH}_3\text{-H}_2\text{O}$**
- 2. $\text{LiBr-H}_2\text{O}$**

OBJECTIVES:

1. To study different components and working of Vapor Absorption Refrigeration System.
2. To compare the cycle with VCRS.
3. Study of actual vapor absorption system and its components.

THEORY:

Some liquids like water have great affinity for absorbing large quantities of certain vapors (NH_3) and reduce the total volume greatly. The absorption refrigeration system differs fundamentally from vapour compression system only in the method of compressing the refrigerant. An absorber, generator and pump in the absorption refrigerating system replace the compressor of a vapour compression system. Ammonia vapour is produced in the generator at high pressure from the strong solution of NH_3 by an external heating source.

The water vapour carried with ammonia is removed in the rectifier and only the dehydrated ammonia gas enters into the condenser. High pressure NH_3 vapour is condensed in the condenser. The cooled NH_3 solution is passed through a throttle valve and the pressure and temperature of the refrigerant are reduced below the temperature to be maintained in the evaporator. The low temperature refrigerant enters the evaporator and absorbs the required heat from the evaporator and leaves the evaporator as saturated vapour.

Slightly superheated, low pressure NH_3 vapour is absorbed by the weak solution of NH_3 which is sprayed in the absorber. In weak NH_3 solution (aqua-ammonia) entering the absorber becomes strong solution after absorbing NH_3 vapour and then it is pumped to the generator through the heat exchanger.

The pump increases the pressure of the strong solution to generator pressure. The strong NH_3 solution coming from the absorber absorbs heat from high temperature weak NH_3 solution in the heat exchanger. The solution in the generator becomes weak as NH_3 vapour comes out of it. The weak high temperature ammonia solution from the generator is passed to the heat exchanger through the throttle valve.

The pressure of the liquid is reduced to the absorber pressure by the throttle valve. $\text{LiBr-H}_2\text{O}$ system is somewhat different than aqua-ammonia based VARS. Water-Lithium Bromide

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(H₂O-LiBr) system is for above 0°C applications such as air conditioning. Here water is the refrigerant and lithium bromide is the absorbent.

QUESTIONS:

1. State the working principle of vapor absorption refrigeration system with neat sketch.
2. Draw the schematic diagram of simple vapor absorption refrigeration system and state its limitations.
3. Draw schematic diagram of actual aqua-ammonia refrigeration system and explain.
4. Explain LiBr-H₂O system with neat sketch. And state its advantages/applications over aqua-ammonia.
5. In a vapor absorption system, the heat is supplied to the generator by condensing steam by at 3 bar and 85% dry. The temperature in the evaporator is to be maintain at -10°C. If the cooling water reject heat at 30°C in a condenser, find the maximum C.O.P. of the system.
6. In a 100 TR aqua ammonia abortion plant, saturated liquid ammonia at 30°C leaves the condenser and enter the expansion valve. The evaporator pressure is 1.9 bar and the vapor temperature exit is -10°C. The mass concentration of ammonia in the weak and strong solution are 0.25 and 0.325 respectively. Determine the mass flow rate (in kg/min) of the strong and weak solutions.
7. The following data refers to a lithium bromide water absorption refrigeration system.
Capacity = 2TR; Concentration of LiBr and enthalpy values for weak solution leaving generator = .68 and 21 kJ/kg respectively. Temperature of water leaving condenser = 40°C; enthalpy of steam leaving evaporator = 2508 kJ/kg; Specific heat of water = 4.2 kJ/kg K. Determine the mass flow rates of strong and weak solutions in kg/min and the heat transfer rates in generator and absorber in kJ/min.

CONCLUSION:

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Refrigeration and Air-Conditioning (ME307)

Date:

Experiment No 5
Psychrometric terms and Processes.

AIM: TO STUDY OF DIFFERENT PSYCHROMETRIC TERMS AND PROCESSES.

OBJECTIVES:

1. To understand different terms related with air properties.
2. Importance of different temperatures for the Psychrometric process.
3. Study of different Psychrometric processes like
 - a) Sensible cooling/heating
 - b) Latent cooling/Heating
 - c) Cooling and dehumidification
 - d) Heating and humidification
 - e) Evaporative cooling

THEORY:

In the design and analysis of air conditioning plants, the fundamental requirement is to identify the various processes being performed on air. Once identified, the processes can be analyzed by applying the laws of conservation of mass and energy. All these processes can be plotted easily on a Psychrometric chart. This is very useful for quick visualization and also for identifying the changes taking place in important properties such as temperature, humidity ratio, enthalpy etc. The important processes that air undergoes in a typical air conditioning plant are discussed below.

Important Psychrometric processes:

a) Sensible cooling:

During this process, the moisture content of air remains constant but its temperature decreases as it flows over a cooling coil. For moisture content to remain constant, the surface of the cooling coil should be dry and its surface temperature should be greater than the dew point temperature of air. If the cooling coil is 100% effective, then the exit temperature of air will be equal to the coil temperature. However, in practice, the exit air temperature will be higher than the cooling coil temperature.

b) Sensible heating

During this process, the moisture content of air remains constant and its temperature increases as it flows over a heating coil.

c) Cooling and dehumidification

When moist air is cooled below its dew-point by bringing it in contact with a cold surface, some of the water vapor in the air condenses and leaves the air stream as liquid, as a result both the temperature and humidity ratio of air decreases as shown. This is the process air undergoes in a typical air conditioning system. Although the actual process path will vary depending upon the type of cold surface, the surface temperature, and flow conditions, for simplicity the process line is assumed to be a straight line. The heat and mass transfer rates can be expressed in terms of the initial and final conditions by applying the conservation of mass and conservation of energy equations.

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d) Heating and Humidification.

During winter it is essential to heat and humidify the room air for comfort. This is normally done by first sensibly heating the air and then adding water vapour to the air stream through steam nozzles.

e) Cooling & humidification.

As the name implies, during this process, the air temperature drops and its humidity increases, this can be achieved by spraying cool water in the air stream. The temperature of water should be lower than the dry-bulb temperature of air but higher than its dew-point temperature to avoid condensation ($T < T_{DPT} < T_O$).

f) Heating and de-humidification

This process can be achieved by using a hygroscopic material, which absorbs or adsorbs the water vapour from the moisture. If this process is thermally isolated, then the enthalpy of air remains constant, as a result the temperature of air increases as its moisture content decreases. This hygroscopic material can be a solid or a liquid. In general, the absorption of water by the hygroscopic material is an exothermic reaction, as a result heat is released during this process, which is transferred to air and the enthalpy of air increases.

QUESTIONS:

1. Explain the following terms in brief. DBT, WBT, DPT
2. Explain the different Psychrometric processes by Psychrometric chart.
3. The atmospheric conditions of air are 35°C dry bulb temperature, 60% relative humidity and 1.01325 bar pressure. If 0.005 kg of moisture per kg of dry air is removed, the temperature becomes 25°C. Determine the final relative humidity and dew point temperature.
4. The atmospheric air at 15°C dry bulb temperature and 80% relative humidity is supplied to the heating chamber at the rate of 100 m³/min. The leaving air has temperature of 22°C without change in its moisture content. Determine the heat added to the air per min and final relative humidity of the air.
5. 100m³ of air at 35°C and 70% relative humidity is cooled to 20°C and 55% relative humidity by passing it through a cooling coil. Using psychrometric tables only, find: 1. The amount of water vapour removed in kg/h; and 2. The cooling capacity of the coil in tonnes of refrigeration. Assume specific heat of superheated vapour in air as 1.88 kJ/kg K and R for air = 287.14 J/kg K.
6. The outside air at 31°C dry bulb temperature and 18.5°C wet bulb temperature enters a cooling coil at the rate of 40 m³/min. the effective surface temperature of the cooling coil is 4.5°C and its cooling capacity is 12.5kW of refrigeration. Find (a) dry bulb temperatures of the air leaving the coil, (b) enthalpy of air leaving the coil, (c) by pass factor of the coil.

CONCLUSION:

Marks obtained:	Signature of faculty:	Date:

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Refrigeration and Air-Conditioning (ME307)

Date:

Experiment No 6
General Cycle Air Conditioning Trainer

THEORY

Air conditioning is making the environment conditions suitable for various applications, like to produce comfortable conditions at home, or to make more complete control of manufacturing processes in industries etc. a complete air conditioning system cleans the air, cools in summer and heats in winter, humidifies in winter and dehumidifies in summer and circulates it through the space where it is required. All the systems may not be equipped with all these controls. This unit is designed to study the basic processes of heating, cooling dehumidification and humidification.

THE APPARATUS

The unit consists of a duct, in which air flow is generated by an axial flow fan. A cooling coil, air heaters, and a steam injector are fitted in the duct. Adjustable flappers are provided in duct to control the air flow. At reduced flow rates of air, cooling coil can be used as dehumidifier. The condensing unit and steam generator is located below the air flow duct. The hermetic compressor compresses the refrigerant vapour and sends it to condenser, where it is liquefied. The liquid refrigerant is throttled by expansion valve to low pressure and temperature. This low temperature refrigerant goes to evaporator where it boils by collecting heat. The low pressure refrigerant vapour coming from the evaporator is sucked by the compressor and again circulated through the system.

A steam generator with electrical heater is provided below the duct, which connected to steam injector in the duct. The steam injection is controlled by a valve. Various measurements are provided so different processes of air conditioning can be studied.

SPECIFICATIONS

Compressor – 0.6 ton (approx.) capacity, hermetically sealed, using refrigerant R-22 (Monochlorodifluoromethane).

Finned tube condenser with fan.

Capillary tube.

Finned tube direct expansion evaporator fitted in a duct of cross section of 250 x 250 mm.

Fan to generate the air flow over the evaporator coil.

Steam generator with electrical heater and steam injector.

Heaters in the duct.

Measurement and controls.

Thermometers for cooling cycle and dry/wet bulb air temperatures.

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Energy meter's for compressor and heater input-02 nos.

Pressure gauges to measure condensing and evaporating pressures.

High/low pressure output.

Necessary switches.

Flow control flappers in the duct.

TERMINOLOGY

Dry air- It is a mixture of constituent gases which comprise the atmospheric air, excluding water vapour.

Moist air- It is the mixture of dry air and water vapour. Generally atmospheric air is moist air.

Humidity – Water vapour that is mixed with atmospheric air is called humidity.

Specific Humidity- It is mass of water vapour in moist air, expresses in Kg / Kg of dry air.

Absolute Humidity – It is mass of water vapour contained in a volume of air and is expressed in grams per m³ of air.

Relative Humidity – It is the ratio of weight of water vapour contained in air at given temperature in a given space to the maximum amount (mass) of water vapour that the air would contain at the same temperature and volume, when fully saturated.

Saturated Vapour – When air – water vapour mixture contains maximum amount of water vapour it can hold, it is called saturated. When temperature of mixture is above saturation temperature, the vapour is supersaturated.

Dalton's Law – In a mechanical mixture of gases, total pressure of the mixture is the sum of partial pressures of gases that each gas would exert when occupied the same volume at the same temperature as that of the mixture.

Dry Bulb Temperature – It is the temperature indicated by ordinary thermometer.

Wet Bulb Temperature – It is temperature indicated by a thermometer whose bulb is covered by a wet wick. The difference between Dry Bulb Temperature and Wet Bulb Temperature is called WET BULB DEPRESSION.

EXPERIMENTAL PROCEDURE

(A) COOLING OF AIR

Fill up water in the wells of DB / WB thermometers.

Adjust the flapper opening to a specific degree. Say 10°C

Put ON main switch. Start air flow through the duct, start the condenser fan and then start the compressor.

Wait till steady temperatures are reached and note down the observations.

Repeat the procedure by changing the air flow.

(B) HEATING OF AIR

Put ON the mains switch and start air flow through the duct.

Put ON the heaters, as required.

Wait till steady state conditions are reached note down the readings.

Repeat the procedure at different air flow rates and changing the heat input.

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(C) HUMIDIFICATION OF AIR

- (1) Fill up sufficient water in steam generator, and start the steam heater.
- (2) After some time, steam will be generated. Now start air flow through the duct, and slightly open the steam control valve so that steam will be injected in the stream of air.
- (3) Note down the readings.

(D) DEHUMIDIFICATION OF AIR

- (1) Start the air flow in the duct. Reduce the flow by partially closing the flap (close the flappers to around half opening.)
- (2) Start the condenser fan and then start the compressor. Cooling coil now works as dehumidifier. Condensed water will start collecting in collecting tray.
- (3) Once steady temperatures are reached, drain all the condensed water. Then again start collecting the condensate.
- (4) Collect the condensate for a period of 15 minutes.
- (5) Note down the temperatures.
- (6) Repeat the procedure by changing air flow.

OBSERVATIONS

(A) COOLING OF AIR

Sr No	Inlet °C		Outlet °C		Flapper	Time for 10 rev. of compressor energy meter	Flow (LPH)
	DBT	WBT	DBT	WBT			

COOLING CYCLE OBSERVATIONS

TEMPERATURES

- (a) Condenser Inlet t_{ci} = °C
(b) Condenser Outlet t_{co} = °C
(c) Evaporator Inlet t_{ei} = °C
(d) Evaporator Outlet t_{eo} = °C

PRESSURES

- (a) Condensing pressure = kg / cm²
(b) Evaporating pressure = kg / cm²
(c) Flapper opening angle = °

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(B) HEATING OF AIR

Sr No	Air inlet °C		Air outlet °C		Flapper	Time for 10 rev. of compressor energy meter
	DBT	WBT	DBT	WBT		

(C) HUMIDIFICATION OF AIR

Sr No	Air inlet °C		Air outlet °C		Flapper opening angle
	DBT	WBT	DBT	WBT	

(D) DEHUMIDIFICATION OF AIR

Sr No	Air inlet °C		Air outlet °C		Flapper opening angle	Condensate collected in 15 min
	DBT	WBT	DBT	WBT		

CALCULATION:

COOLING OF AIR

Inlet – DBT = _____ °C, WBT = _____ °C
 Therefore, from table of Dry / Wet Bulb Temperatures,
 Relative humidity,
 R H = _____ %

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$$\text{Now, } RH = \frac{P_w}{P_{sat}} \times 100$$

Where, P_w = Partial pressure of water vapour, Kg / cm²

P_{sat} = Partial pressure of water vapour

At saturation at DBT (to be taken from chart)

$$\therefore P_w = \frac{R_H \times P_{sat}}{100} \quad \text{Kg / cm}^2$$

Now,

$$W = \frac{0.623 \times P_w}{(P - P_w)}$$

Where, W = weight of moisture Kg/Kg of dry air

P = Atmospheric pressure Kg / cm²

(1) TOTAL ENTHALPY OF DRY AIR = $H_t = h_a + h_s + h_l + h_{sh}$

Where, h_a = Enthalpy of dry air = $1 \times (\text{DBT}) \text{ KJ / Kg}$

h_s = Sensible heat of water vapour (up to WBT) = $4.2 \times w \times (\text{WBT})$

h_l = Latent heat of evaporation of water vapour (at wbt) = $w \times l$
 (l is taken from chart)

h_{sh} = Superheat of water vapour (from WBT to DBT)
 = $w \times 1.9 \times (\text{DBT} - \text{WBT})$

Thus, calculate total heat of air at inlet and outlet ie, h_{ti} and h_{to} respectively.

(2) AIR FLOW

(See table-2, showing air velocity ' V_a ' at a corresponding flapper opening)

Cross sectional area of duct 0.0529 m²

$$\text{Density of air, } \rho_a = \frac{1.293 \times 273}{273 + \text{DBT}_{out}}$$

So, mass flow rate of air

$$m_a = (0.0529 - 0.0188) \times V_a \times \rho_a \times 3600 \quad \text{Kg/hr}$$

(3) HEAT RECOVERED BY COOLING COIL

$$H = m_a \cdot (h_{ti} - h_{to}) \quad \text{Kg/hr}$$

(4) COOLING CYCLE PERFORMANCE

(i) Evaporating Pressure, $P_{eg} = \text{_____ kg / cm}^2 \text{ ab}$

so absolute evaporating pressure, $P_{ea} = P_{eg} + 1.033 \text{ kg / cm}^2 \text{ ab}$

(ii) Condensing Pressure, $P_{cg} = \text{_____ kg / cm}^2 \text{ ab}$

so Absolute Condensing pressure, $P_{ca} = P_{cg} + 1.033 \text{ kg / cm}^2 \text{ ab}$

(iii) From the temperatures, plot the cycle on P-H chart and find out enthalpy values.

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$$H_{ci} = \text{_____} \text{ Kj/Kg}$$

$$H_{co} = H_{ei} = \text{_____} \text{ Kj/Kg}$$

$$H_{eo} = \text{_____} \text{ Kj/Kg}$$

$$(vi) \text{ Refrigerating effect} = H_{eo} - H_{ei} = \text{_____} \text{ Kj/Kg}$$

$$(v) \text{ Compressor work} = H_{ci} - H_{eo} = \text{_____} \text{ Kj/Kg}$$

$$(vi) \text{ COP theo} = (H_{eo} - H_{ei}) / (H_{ci} - H_{eo})$$

(vii) If time required for 10 rev of compressor energy meter is t_{com} sec.

Compressor work,

$$C_w = \frac{3600 \times 10}{t_{com} \times EMCC} \text{ Kw}$$

Where, EMCC = Compressor energy meter constant = 3200 imp/kwh

$$(viii) \text{ COP act} = H / C_w$$

(B) HEATING OF AIR

In a similar manner to cooling cycle, calculate heat gained by air $H = m_a (h_{ti} - h_{to})$ Kj/hr

(C) HUMIDIFICATION OF AIR

At inlet conditions, DB = _____ °C, WB = _____ °C, So RH_i = _____ %

At outlet conditions, DB = _____ °C, WB = _____ °C, So RH_o = _____ %

Increment of relative humidity = RH_o – RH_i

(D) DEHUMIDIFICATION OF AIR

At inlet conditions, DB = _____ °C, WB = _____ °C, So RH = _____ % So W_i = _____ Kg/Kg of air

At outlet conditions, DB = _____ °C, WB = _____ °C, So RH = _____ % So W_o = _____ Kg/Kg of air

Mass flow rate of air = m_a Kg/hr

Moisture removed within 15 minutes

$$m_a = (W_i - W_o) \times 15/60 \text{ Kg/hr}$$

Compare this with actual amount of condensate collected.

Actually, in large capacity year around air conditioning plants, instead of using these Processes individually a combination of processes is also used. Like dehumidification with heating in summer, or humidification with heating in winter. Equipment used for various processes depend upon the resources available, system requirements and accuracy and economy of comfort desired.

Table – 1

THERMODYNAMIC PROPERTIES OF R – 22

TEMP °C	SPECIFIC VOLUME m ³ / kg	
	Sat. liquid	Sat. vapour
-10	0.00075876	0.0653399
-5	0.00076831	0.0553394
0	0.00077834	0.0471354
4	0.00078673	0.0416124
8	0.00079549	0.0368493
12	0.00080465	0.0327239
16	0.00081424	0.0291361

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20	0.00082431	0.0260032
24	0.00083491	0.0232572
26	0.00084043	0.0220111
28	0.00084610	0.0208411
30	0.00085193	0.0197417
32	0.00085793	0.0187076
34	0.00086412	0.0177341
36	0.00087051	0.0168168
38	0.00087710	0.0159517
40	0.00088394	0.0151351

Table-2

FLAPPER CLOSING (DEGREES)	AIR VELOCITY (m/s)
0 (fully open)	5.0
10	1.5
20	2.7
30	3.2
40	3.6
60	4.3
70	4.4
80	4.8

CONCLUSION:

Marks obtained:	Signature of faculty:	Date:

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Refrigeration and Air-Conditioning (ME307)

Date:

Experiment No 7
Air-Conditioning Systems

AIM: To Study different air-conditioning systems

- 1. Window Air-conditioner**
- 2. Split Air-conditioner**
- 3. Dessert Cooler.**
- 4. Air-washer**

OBJECTIVES:

1. To study and understand different commercial air-conditioning systems.
2. To study window A.C and split A.C in house hold applications.
3. Comparison of window A.C and Split A.C.
4. Application and processes of air-washer.

THEORY:

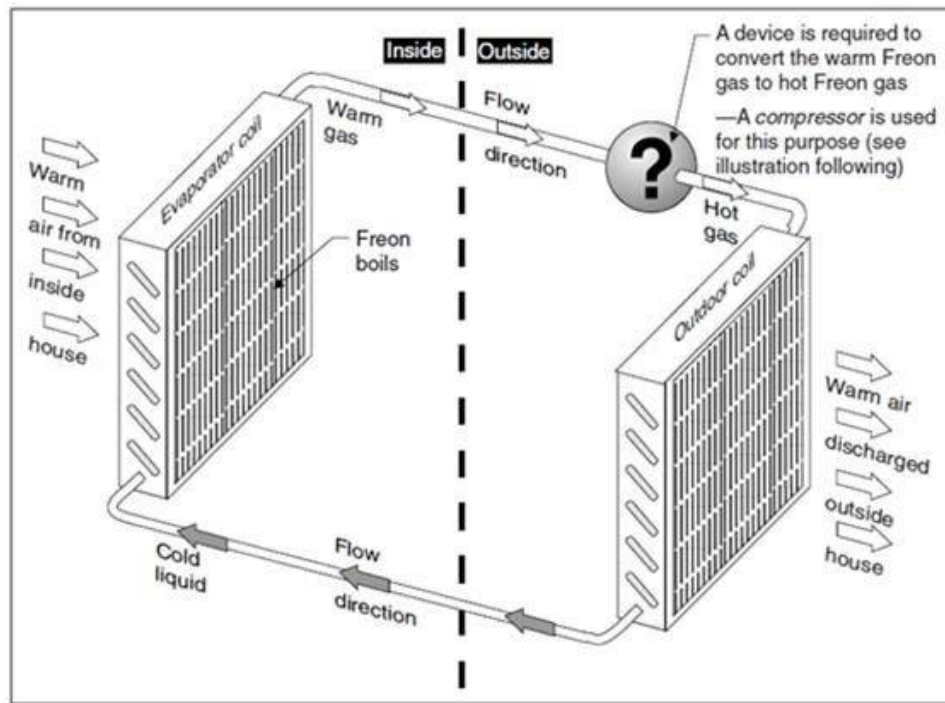
The most common type of air conditioning that we see is technically referred to as direct expansion, mechanical, vapor-compression refrigeration system. Like a Refrigerator The goal with air conditioning is to capture heat in the house and throw it outside. One of the ways we can think about it is to look at a refrigerator. If we can keep the temperature inside your refrigerator at about 40°F and it is 70°F in the kitchen, somehow we are taking heat out of that cool air and dumping it into a kitchen that is warmer.

Central air conditioning and refrigerators operate on exactly the same principle. The process works something like this. The Coils We have two coils similar to the radiator in a car: one inside the house and one outside. We put something cold through the coil inside the house and then blow warm house air across the coil, so the coil can grab heat from the house air. This cools the house. We want to take that heat in the coil outside and dump it into the outdoor air.

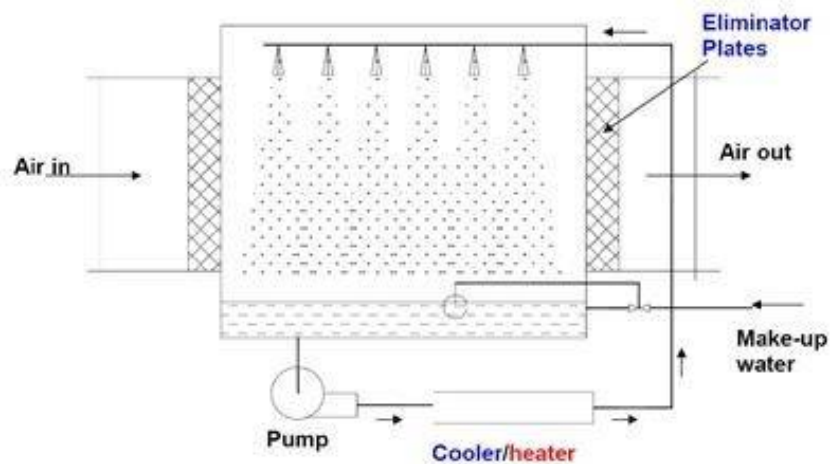
In Split type air-conditioning system, the working principle as well as the main components are same like window air-conditioner but the difference is that the condenser and compressor in split a.c are outside the room in order to make the conditioned room more silent. Only insulated transfer line are provided to handle the refrigerant to transfer from evaporator to condenser. An evaporative cooler (also swamp cooler, desert cooler, and wet air cooler) is a device that cools air through the evaporation of water. Evaporative cooling differs from typical air conditioning systems which use vapor-compression or absorption refrigeration cycles. Evaporative cooling works by employing water's large enthalpy of vaporization.

The temperature of dry air can be dropped significantly through the phase transition of liquid water to water vapor, which requires much less energy than refrigeration. In extremely dry climates, it also has the added benefit of conditioning the air with more moisture for the comfort of occupants. Unlike refrigeration, it requires a water source, and must continually consume water to operate.

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An air washer is a device for conditioning air. As shown in Fig, in an air washer air comes in direct contact with a spray of water and there will be an exchange of heat and mass (water vapour) between air and water. The outlet condition of air depends upon the temperature of water sprayed in the air washer. Hence, by controlling the water temperature externally, it is possible to control the outlet conditions of air, which then can be used for air conditioning purposes.



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CONCLUSION:

QUESTIONS:

1. Explain window air conditioner with neat sketch and Psychrometric process.
2. Explain split air-conditioner with neat sketch.
3. Explain evaporative cooler with neat sketch and Psychrometric process.
4. Explain the working principle of air washer with neat sketch.

Marks obtained:	Signature of faculty:	Date:

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Refrigeration and Air-Conditioning (ME307)

Date:

Experiment No 8

AIM: - Design of air conditioning system and load calculation for residential and commercial building (using CLTD method)

CLTD/SCL/CLF METHOD OF LOAD CALCULATION (ASHRAE FUNDAMENTALS 1997)

As mentioned before, the heat gain to the building is not converted to cooling load instantaneously. CLTD (cooling load temperature difference), SCL (solar cooling load factor), and CLF (cooling load factor): all include the effect of (1) time-lag in conductive heat gain through opaque exterior surfaces and (2) time delay by thermal storage in converting radiant heat gain to cooling load.

This approach allows cooling load to be calculated manually by use of simple multiplication factors.

- a. CLTD** is a theoretical temperature difference that accounts for the combined effects of inside and outside air temp difference, daily temp range, solar radiation and heat storage in the construction assembly/building mass. It is affected by orientation, tilt, month, day, hour, latitude, etc. CLTD factors are used for adjustment to conductive heat gains from walls, roof, floor and glass.
- b. CLF** accounts for the fact that all the radiant energy that enters the conditioned space at a particular time does not become a part of the cooling load instantly. The CLF values for various surfaces have been calculated as functions of solar time and orientation and are available in the form of tables in ASHRAE Handbooks. CLF factors are used for adjustment to heat gains from internal loads such as lights, occupancy, power appliances.
- c. SCL** factors are used for adjustment to transmission heat gains from glass.

1. External Cooling Load

As discussed before, the total cooling load on a building consists of external as well as internal loads. The external loads consist of heat transfer by conduction through the building walls, roof, floor, doors etc, heat transfer by radiation through fenestration such as windows and skylights. All these are sensible heat transfers.

1.1 ROOF

The basic conduction equation for heat gain is $q = U A \Delta T$.

Where

q = Heat gain in Btu/hr

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U = Thermal Transmittance for roof in Btu/hr.ft².°F

A = area of roof in ft²

ΔT = Temperature difference in °F

The heat gain is converted to cooling load using the room transfer functions (sol-air temperature) for the rooms with light, medium and heavy thermal characteristics. The equation is modified as

$Q = U * A * (CLTD)$ Where

Q = cooling load, Btu/hr

U = Coefficient of heat transfer roof or wall or glass, Btu/hr.ft².°F

A = area of roof, ft²

$CLTD$ = cooling load temperature difference °F. The values are determined from tables available in chapter 28 of ASHRAE fundamentals handbook.

Since the ASHRAE tables provide hourly $CLTD$ values for one typical set of conditions i.e. outdoor maximum temperature of 95°F with mean temperature of 85°F and daily range of 21°F, the equation is further adjusted to apply correction factors for conditions other than the mentioned base case. Thus,

$Q_{\text{Roof}} = U * A * CLTD_{\text{Roof Corrected}}$

The typical steps to calculate the Roof load are as follows:

Step #1 Determine roof construction and overall heat transfer coefficient (U) (Chapter 28 ASHRAE 1997, Table A24-4, A29-5)

Step #2 Select roof no. from ASHRAE Table 31 or Text table 7-34 which is closest to matching actual roof construction.

Step #3 Select $CLTD_{\text{Roof}}$ for time of interest, typically on an hourly basis (Chapter 28 ASHRAE Table A28-32, A28-34).

Step #4 Corrections: Values on tables are for

- a. Four latitudes on July or August
- b. Indoor temperature of 78°F
- c. Outdoor maximum temperature of 95°F with mean daily temperature of 85°F and daily range of 21°F

$CLTD_{\text{ROOF Corrected}} = [CLTD_{\text{Roof}} + (78 - TR) + (TM - 85)]$

Where

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$(78 - TR)$ = indoor design temperature correction, $(TM - 85)$ = outdoor design temperature correction, TR = Indoor room temperature

T_m = Mean outdoor temperature

T_{max} = Maximum outdoor temperature o $T_m = T_{max} - (\text{Daily Range}) / 2$

Step # 5 Calculate roof area (A) from architectural plans

Step #6 $Q_{\text{Roof}} = U * A * CLTD_{\text{Roof Corrected}}$

What is Sol-air temperature?

The sol-air temperature is a fictitious number that defines the value of the outside air temperature which would, in the absence of all radiation exchanges, give the same rate of heat flow (Q) into the outer surface of the wall as the actual combination of temperature difference and radiation exchanges.

The sol-air temperature represents the equivalent outdoor design air temperature that combines convection to the outdoor air, radiation to the ground and sky, and solar radiation heat transfer effects on the outer surface of a building. It takes into account the mass of the wall or roof, the daily temperature range (which affects the heat storage), the color of the outside surface (which affects solar heat absorption rate) and the latitude and month. When the sol-air data is combined with the inside design temperature, a cooling load temperature difference (CLTD) is obtained.

Hourly sol-air temperature values are presented in tables for various orientations of a surface for 21 July at 40° N latitude, with standard surface absorption factor for light and dark colored surfaces in ASHRAE handbook of fundamentals. The sol-air temperatures are listed for a given air temperature cycle.

Adjustments can be introduced for other dates, latitudes and air temperature cycles. When this method is used in conjunction with custom tables generated by appropriate computer software, and for buildings where external shading is not significant, it can be expected that it will produce results very close to that produced by the TFM.

1.2 WALLS

The cooling load from walls is treated in a similar way as roof:

$$Q_{\text{Wall}} = U * A * CLTD_{\text{Wall Corrected}}$$

Where

Q_{Wall} = Load through the walls in Btu/hr

U = Thermal Transmittance for walls in Btu/ (h ft² F)

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A = area of walls in ft²

CLTD = Cooling Load Temperature Difference for walls in °F

The typical steps to calculate the Wall load are as follows:

Step #1 Determine wall construction and overall heat transfer coefficient (U) (Chapter 24 ASHRAE 1997, Table A24-4, A29-5)

Step #2 Select wall type from ASHRAE Table 33 which is closest to matching actual wall construction.

Pay attention to effect of mass distribution (inside insulation, outside insulation or evenly distributed).

Step #3 Select CLTD_{wall} for time of interest, typically on an hourly basis (Chapter 28 ASHRAE Table A28-32, A28-34).

Step #4 Corrections: Values on tables are for

- a. Four latitudes on July or August
- b. Indoor temperature of 78°F
- c. Outdoor maximum temperature of 95°F with mean daily temperature of 85°F and daily range of 21°F

CLTD_{wall} Corrected = [CLTD_{wall} + (78 – TR) + (TM – 85)] Where

(78 – TR) = indoor design temperature correction ,

(TM – 85) = outdoor design temperature correction

T_m = Mean outdoor temperature

o T_{max} = Maximum outdoor temperature o T_m = T_{max} – (Daily Range) / 2

Step #5 Calculate walls area (A). Use the architectural drawings to determine how much wall area you have. Since you will be finding window and door losses separately, exclude windows and door and count *only* wall area.

Step #6 **Q_{wall} = U * A * CLTD_{wall} Corrected**

1.3 SOLAR LOAD THROUGH GLASS

Solar load through glass has two components: 1) Conductive and 2) Solar Transmission

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The absorbed and then conductive portion of the radiation through the windows is treated like the roof & walls where CLTD values for standard glazing are tabulated in ASHARE fundamentals handbook.

For solar transmission, the cooling load is calculated by the cooling load SCL factor and shading coefficient (SC).

The cooling load equations for glass are:

Conductive $Q_{\text{Glass Conductive}} = U * A * \text{CLTD}_{\text{Glass Corrected}}$

Solar Transmission $Q_{\text{Glass Solar}} = A * SC * SCL$

Where

$Q_{\text{Conductive}}$ = Conductive load through the glass in Btu/hr

Q_{Solar} = Solar transmission load through the glass in Btu/hr

U = Thermal Transmittance for glass in Btu/ (h ft² F)

A = area of glass in ft²

CLTD = Cooling Load Temperature Difference for glass in °F

SC = Shading coefficient

SCL = Solar Cooling Load Factor

STEPS FOR CONDUCTIVE CALCULATIONS

Step # 1 For the glass types used, select from ASHRAE tables the overall heat transfer coefficient (U). The glass is the major contributor of heat gain in the commercial buildings. Pay attention to effect of shading, reflective films, curtains, drapes etc (refer to example below)

Step # 2 Select $\text{CLTD}_{\text{Glass}}$ for time of interest, typically on an hourly basis (Chapter 28 ASHRAE Table 34).

Step # 3 Corrections:

$\text{CLTD}_{\text{Glass Corrected}} = [\text{CLTD}_{\text{Glass}} + (78 - \text{TR}) + (\text{TM} - 85)]$ Where

o (78 - TR) = indoor design temperature correction o (TM - 85) = outdoor design temperature correction o TR = Indoor room temperature

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o T_m = Mean outdoor temperature

o T_{max} = Maximum outdoor temperature o $T_m = T_{max} - (\text{Daily Range}) / 2$

Step # 4 Calculate glass area (A) from architectural plans

Step # 5 $Q_{\text{Glass}} = U * A * CLTD_{\text{Glass Corrected}}$

STEPS FOR SOLAR TRANSMISSION CALCULATIONS

Step # 1 Determine shading coefficient (SC) from ASHRAE 1997 Chapter 27, Table 11

Step # 2 Determine zone type from ASHRAE 1997 Chapter 29, Table 35 B

Step # 3 Determine solar cooling load factor (SCL) from ASHRAE 1997 Chapter 28, Table A28-36

Step # 4 Calculate glass area (A) from architectural plans

Step # 5 $Q_{\text{Glass Solar}} = A * SC * SCL$

1.4 PARTITIONS, CEILINGS AND FLOORS

Whenever a conditioned space is adjacent to a space with a different temperature, transfer of heat through the separating physical section must be considered.

$Q = U A (T_a - T_{rc})$ Where

U = coefficient of overall heat transfer between adjacent and conditioned space in Btu/ (h ft² F)
See 1997 ASHRAE Fundamentals, Chapter 24 or 2001 ASHRAE Fundamentals, and Chapter 25.

A = area of partition in ft², ceiling or floor calculated from building plans

T_a = Temperature of adjacent space in °F (Note: If adjacent space is not conditioned and temperature is not available, use outdoor air temperature less 5°F)

T_{rc} = Inside design temperature of conditioned space in °F (assumed constant)

Note that the temperature T_a may range widely from that in the conditioned space. The temperature in a kitchen or boiler room, for example, may be as much as 15 to 50°F above the outdoor air temperature. Actual temperatures in adjoining spaces should be measured when possible. Where nothing is known, assume outdoor air temperature less 5°F.

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2 Internal Cooling Loads

The various internal loads consist of sensible and latent heat transfers due to occupants, products, processes appliances and lighting. The lighting load is only sensible. The conversion of sensible heat gain (from lighting, people, appliances, etc.) to space cooling load is affected by the thermal storage characteristics of that space and is thus subject to appropriate cooling load factors (CLF) to account for the time lag of the cooling load caused by the building mass. The weighting factors equation determines the CLF factors.

$$\text{CLF} = \text{Q cooling load} / \text{Q internal gains}$$

Note that the latent heat gains are considered instantaneous.

2.1 People

$$\text{Q sensible} = N (\text{Q}_s) (\text{CLF})$$

$$\text{Q latent} = N (\text{Q}_L) \text{ Where}$$

- N = number of people in space from ASHRAE, Table A28-3

Q_s , Q_L = Sensible and Latent heat gain from occupancy is given in 1997 ASHRAE Fundamentals Chapter 28, Table 3)

CLF = Cooling Load Factor, by hour of occupancy. See 1997 ASHRAE Fundamentals, Chapter 28, Table 37

Note: CLF = 1.0, if operation is 24 hours or of cooling is off at night or during weekends.

The table below gives representative rates at which heat and moisture are given off by human beings in different states of activity. Often these sensible and latent heat gains constitute a large fraction of the total load. Even for short-term occupancy, the extra heat and moisture brought in by people may be significant.

Snapshot of Occupancy Heat gain values in ASHRAE

Heat Gain from Occupants at Various Activities (At Indoor Air Temperature of 78°F)

Activity	Total heat, Btu/h		Sensible heat, Btu/h	Latent heat, Btu/h
	Adult, male	Adjusted		
Seated at rest	400	350	210	140
Seated, very light work, writing	480	420	230	190
Seated, eating	520	580	255	325
Seated, light work, typing,	640	510	255	255
Standing, light work or walking	800	640	315	325
slowly,	880	780	345	435
Light bench work	1040	1040	345	695
Light machine work, walking 3mi/hr	1360	1280	405	875
Moderate dancing	1600	1600	565	1035

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2.2 Lights

The primary source of heat from lighting comes from light-emitting elements. Calculation of this load component is not straightforward; the rate of heat gain at any given moment can be quite different from the heat equivalent of power supplied instantaneously to those lights. Only part of the energy from lights is in the form of convective heat, which is picked up instantaneously by the air-conditioning apparatus. The remaining portion is in the form of radiation, which affects the conditioned space only after having been absorbed and re-released by walls, floors, furniture, etc. This absorbed energy contributes to space cooling load only after a time lag, with some part of such energy still present and reradiating after the lights have been switched off.

Generally, the instantaneous rate of heat gain from electric lighting may be calculated from: $Q = 3.41 \times W \times \text{FUT} \times \text{FSA}$

Cooling load factors are used to convert instantaneous heat gain from lighting to the sensible cooling load; thus the equation is modified to

$Q = 3.41 \times W \times \text{FUT} \times \text{FSA} \times (\text{CLF})$, Where

W = Watts input from electrical lighting plan or lighting load data

FUT = Lighting use factor, as appropriate

FSA = special ballast allowance factor, as appropriate

CLF = Cooling Load Factor, by hour of occupancy. See 1997 ASHRAE Fundamentals, Chapter 28, Table 38

Note: CLF = 1.0, if operation is 24 hours or if cooling is off at night or during weekends.

2.3 Power Loads

The industrial and commercial applications use various equipments such as fans, pumps, machine tools, elevators, escalators and other machinery, which add significantly to the heat gain. There are 5 equations in use for different scenarios.

Case #1

If the motor and the machine are in the room the heat transferred can be calculated as

$Q = 2545 \times (P / \text{Eff}) \times \text{FUM} \times \text{FLM}$

P = Horsepower rating from electrical power plans or manufacturer's data

Eff = Equipment motor efficiency, as decimal fraction

FUM = Motor use factor (normally = 1.0)

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FLM = Motor load factor (normally = 1.0)

Note: FUM = 1.0, if operation is 24 hours

In this situation the total power are transferred as heat to the room.

Note! If the machine is a pump or a fan most of the power are transferred as energy to the medium and may be transported out of the room. For such cases, case 4 shall be used.

Case #2

If the motor is outside and the machine is in the room the heat transferred can be calculated as

$$Q = 2545 * P * FUM * FLM$$

P = Horsepower rating from electrical power plans or manufacturer's data

FUM = Motor use factor

FLM = Motor load factor

Note: FUM = 1.0, if operation is 24 hours

Case #3

If the motor is belt driven and the motor and belt is outside and the machine is in the room the heat transferred can be calculated as

$$Q = 2545 * P * \text{Belt Eff} * FUM * FLM$$

P = Horsepower rating from electrical power plans or manufacturer's data

Belt Eff = Belt transmission efficiency, as decimal fraction

FUM = Motor use factor

FLM = Motor load factor

Note: FUM = 1.0, if operation is 24 hours

Case #4

If the motor is in the room and the machine is outside the heat transferred can be calculated as

$$Q = 2545 * [P/\text{Eff} - P] * FUM * FLM$$

P = Horsepower rating from electrical power plans or manufacturer's data

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Eff = Motor efficiency, as decimal fraction

FUM = Motor use factor

FLM = Motor load factor

Note: FUM = 1.0, if operation is 24 hours

Case #5

If the motor is belt driven and the motor and belt is in the room and the machine is outside the heat transferred can be calculated as

$$Q = 2545 * [P / (\text{motor Eff}) - P / (\text{belt Eff})] * FUM * FLM$$

P = Horsepower rating from electrical power plans or manufacturer's data FUM = Motor use factor, FLM = Motor load factor

2.4 Appliances

In a cooling load estimate, heat gain from all appliances-electrical, gas, or steam-should be taken into account. Because of the variety of appliances, applications, schedules, use, and installations, estimates can be very subjective. Often, the only information available about heat gain from equipment is that on its name-plate.

$$Q_{\text{Sensible}} = Q_{\text{in}} \times F_u \times F_r \times (\text{CLF})$$

$$Q_{\text{Latent}} = Q_{\text{in}} \times F_u \text{ Where}$$

Q_{in} = rated energy input from appliances. See 1997 ASHRAE Fundamentals, Chapter 28, and Table 5 thru 9 or use manufacturer's data. For computers, monitors, printers and miscellaneous office equipment, see 2001 ASHRAE Fundamentals, Chapter 29, Tables 8, 9, & 10. [If the appliance nameplate provides the power ratings in Watts, Q_{in} can be estimated as $3.14 \times \text{Wattage of equipment}$]

F_u = Usage factor. See 1997 ASHRAE Fundamentals, Chapter 28, Table 6 and 7

F_r = Radiation factor. See 1997 ASHRAE Fundamentals, Chapter 28, Table 6 and 7

CLF = Cooling Load Factor, by hour of occupancy. See 1997 ASHRAE Fundamentals, Chapter 28, Table 37 and 39

Note 1: CLF = 1.0, if operation is 24 hours or of cooling is off at night or during weekends.

Note 2: Set latent load = 0 if appliance under exhaust hood.

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2.5 Infiltration Air

$$Q_{\text{sensible}} = 1.08 \times \text{CFM} \times (T_o - T_i) \quad Q_{\text{latent}} = 4840 \times \text{CFM} \times (W_o - W_i)$$

$$Q_{\text{total}} = 4.5 \times \text{CFM} \times (h_o - h_i)$$

Where

CFM = Infiltration air flow rate. See 1997 ASHRAE Fundamentals, Chapter 25, for determining infiltration

T_o, T_i = Outside/Inside dry bulb temperature, °F

W_o, W_i = Outside/Inside humidity ratio, lb water/lb dry air

h_o, h_i = Outside/Inside air enthalpy, Btu per lb (dry air)

3 Heat Gain from Miscellaneous Sources

3.1 Supply Fan Heat Load

Supply and/or return fans that circulate or supply air to the space add heat to the space or system depending on the location relative to the conditioned space. The heat added may take one or all of the following forms:

Instantaneous temperature rise in the air stream due to fan drive inefficiency.

Temperature rise in the air stream when the air is brought to static equilibrium and the static and kinetic energy is transformed into heat energy.

The location of the fan and motor relative to the cooling coil and space being conditioned determines how the heat is added to the system. If the fan is downstream of the cooling coil (draw-thru) then the fan heat load is added to the space cooling load. If the fan is upstream of the cooling coil, then the fan heat load is added to the system cooling coil load.

The heat energy is calculated as follows:

$$Q = 2545 \times [P / (\text{Eff1} \times \text{Eff2})] \text{ Where}$$

P = Horsepower rating from electrical power plans or manufacturer's data

2545 = conversion factor for converting horsepower to Btu per hour

Eff1 = Full load motor and drive efficiency,

Eff2 = Fan static efficiency

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3.2 Ventilation Air

Ventilation air is the amount of outdoor air required to makeup for air leaving the space due to equipment exhaust, exfiltration and/or as required to maintain Indoor Air Quality for the occupants. (See ASHRAE Standard 62 for minimum ventilation requirements). The heat is usually added to the air stream before the cooling coil and has no direct impact on the space conditions. The additional cooling coil load is calculated as follows:

$$Q_{\text{sensible}} = 1.08 \times \text{CFM} \times (T_o - T_c),$$

$$Q_{\text{latent}} = 4840 \times \text{CFM} \times (W_o - W_c)$$

$$Q_{\text{total}} = 4.5 \times \text{CFM} \times (h_o - h_c)$$

Where

CFM = Ventilation airflow rate.

T_o = Outside dry bulb temperature, °F

T_c = Dry bulb temperature of air leaving the cooling coil, °F

W_o = Outside humidity ratio, lb (water) per lb (dry air)

W_c = Humidity ratio of air leaving the cooling coil, lb (water) per lb (dry air)

h_o = Outside/Inside air enthalpy, Btu per lb (dry air)

h_c = Enthalpy of air leaving the cooling coil Btu per lb (dry air)

3.3 Duct Heat Gain

Unless the return ductwork system is extensive and uninsulated or passes over a non-conditioned space, only the heat gained by the duct supply system is significant. This heat gain is normally estimated as a percentage of the space sensible cooling load (usually 1% to 5%) and applied to the temperature of the air leaving the cooling coil in the form of temperature increase.

3.4 Duct Leakage

Air leakage out of or into ductwork can have much greater impact than the duct heat gain or loss. Outward leakage from a supply duct is a direct loss of cooling and/or dehumidifying capacity and must be offset by increased air flow or reduced supply air temperature. Leakage into a return duct system causes additional cooling coil capacity but it does not directly affect the space conditions.

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Commercial type or existing older systems can have leakage from **10% to 20%** of the total system airflow. Per energy conservation guidelines, in a new installation, the duct system should not leak more than **1% to 3%** of total system airflow.

The engineer or designer is cautioned to make sure that the proper allowance for leakages is included in the calculations in order to ensure that the equipment selected is properly sized.

3.5 Diversity Factors

Diversity of cooling load results from not using part of the load on a design day. Therefore, diversity factors are factors of usage and are applied to the refrigeration capacity of large air conditioning systems. These factors vary with location, type and size of application. Generally, diversity factors can be applied to loads from people and lights as there is neither 100% occupancy nor total lighting at the time of such other peak loads as peak solar and transmission loads. The reduction in cooling loads from non-use is real and should be accounted for. Listed are some average typical diversity factors for large buildings during occupied periods.

Type of Application	Diversity Factor	
	Lights	People
Office	0.70 to 0.85	0.75 to 0.90
Apartment, Hotel	0.30 to 0.50	0.40 to 0.60
Department Store	0.90 to 1.00	0.80 to 0.90
Industrial	0.80 to 0.90	0.85 to 0.95

In the case of Industrial, diversity should also be applied to the machinery load.

4. SUPPLY AIR CALCULATIONS

Calculations for the design supply air quantities are dependent on the type of system used. Some generalizations and assumptions need to be made to assist in the use of the cooling load calculations for equipment selection and sizing. For constant air volumes with reheat type of system, the design supply air quantities required are based on the peak requirements for each space. However, if the system selected allows for air flow to each zone to vary based on load, the peak load on an air conditioning unit serving several zones or spaces is not equal to the sum of the peak loads of each zone, but will be that amount necessary to handle the maximum coincident load on the

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system at a given time. Determining the maximum coincident load may require additional calculation and analysis.

Supply air flow rate to a space is based only on the total space sensible heat load, thus **CFM = $1.08 \times [Q_{\text{sensible}} / (TR - T_s)]$**

Where

CFM = air flow in cubic feet per minute

1.08 = conversion constant = $0.244 \times (60/13.5)$

0.244 = specific heat of moist air, Btu/lb d.a.

13.5 = specific volume of moist air, cu-ft. per lb. d.a. (@70°F, 50% RH)

Q sensible = total room sensible heat gain, BTU per hr.

TR = Room dry bulb temperature, °F

T_s = Room supply air dry bulb temperature, °F (not necessarily the same as the temperature of the air leaving the cooling coil).

CONCLUSION:

Marks obtained:	Signature of faculty:	Date:

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Refrigeration and Air-Conditioning (ME307)

Date:

Experiment No 9

Study of Sling Psychrometer

AIM: To study psychometric terms and sling psychrometer.

OBJECTIVES:

To measure DBT and WBT at different locations.

OBSERVATION TABLE:

Sr No	DBT	WBT

COMPUTATION:

Using Psychrometric chart calculate

Partial pressure of water vapour

Relative humidity

Density of water vapour

Dew point temperature

Degree of saturation

Partial pressure of air

Density of Air

Specific Humidity

Enthalpy of Mixture

Specific volume of air

Verify results by Calculation also.

QUESTIONS:

1. Define DBT, WBT, DPT
2. Define: Specific Humidity, Percentage Humidity and Relative Humidity. Also prove relation between relative humidity and percentage humidity.
3. Explain with neat sketch the sling psychrometer.

CONCLUSION:

Marks obtained:	Signature of faculty:	Date:

Refrigeration and Air-Conditioning (ME307)

Date:

Experiment No 10

Exhaust gas heat recovery type VAR system

Introduction:

Refrigeration is the process of removing heat from an enclosed or controlled space, or from a substance, and moving it to a place where it is unobjectionable. The primary purpose of refrigeration is lowering the temperature of the enclosed space or substance and then maintaining that lower temperature as compare to surroundings. Cold is the absence of heat, hence in order to decrease a temperature, one "removes heat", rather than "adding cold." The basic objective of developing a vapour absorption refrigerant system for cars is to cool the space inside the car by utilizing waste heat and exhaust gases from engine.

The air conditioning system of cars in today's world uses "Vapour Compression Refrigerant System" (VCRS) which absorbs and removes heat from the interior of the car which is the space to be cooled and further rejects the heat to be elsewhere. Now to increase an efficiency of car beyond a certain limit vapour compression refrigerant system resists it as it cannot make use of the exhaust gases from the engine. In vapour compression refrigerant system, the system utilizes power from engine shaft as the input power to drive the compressor of the refrigerant system, hence the engine has to produce extra work to run the compressor of the refrigeration system utilizing extra amount of fuel. This loss of power of the vehicle for refrigeration can be neglected by utilizing another refrigeration system i.e. a "Vapour Absorption Refrigerant System" i.e. low grade heat operated systems.

It is well known that an IC engine has an efficiency of about 35-40%, which means that only one-third of the energy in the fuel is converted into useful work and about 60-65% is wasted to environment. In which 28-30% is lost by cooling water and lubrication losses, around 30-32% is lost in the form of exhaust gases and remainder by radiation, etc. In a Vapour Absorption Refrigerant System, a physicochemical process replaces the mechanical process of the Vapour Compression Refrigerant System by using energy in the form of heat rather than mechanical work. The heat required for running the Vapour Absorption Refrigerant System can be obtained from that which is wasted into the atmosphere from IC engine. Hence to utilize the exhaust gases and waste heat from an engine the vapour absorption refrigerant system can be put into practice which increases the overall efficiency of a car.

Possibility of heat recovery and availability from I.C. engine

Waste heat is heat, which is generated in a process by way of fuel combustion or chemical reaction, and then "dumped" into the environment even though it could still be reused for some useful and economic purpose. This heat depends in part on the temperature of the waste heat gases and mass flow rate of exhaust gas. Waste heat losses arise both from equipment inefficiencies and from thermodynamic limitations on equipment and processes. For example, consider internal combustion engine approximately 30 to 40% is converted into useful mechanical work.

The remaining heat is expelled to the environment through exhaust gases and engine cooling systems. It means approximately 60 to 70% energy losses as a waste heat through exhaust (30%

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as engine cooling system and 30 to 40% as environment through exhaust gas). Exhaust gases immediately leaving the engine can have temperatures as high as 842-1112°F [450- 600°C]. Consequently, these gases have high heat content, carrying away as exhaust emission. Efforts can be made to design more energy efficient reverberatory engine with better heat transfer and lower exhaust temperatures; however, the laws of thermodynamics place a lower limit on the temperature of exhaust gases.

Working principle: Vapour absorption refrigeration system

The vapour-absorption cycle is similar to the better known vapour-compression cycle in that it employs a volatile refrigerant, e.g. ammonia, which alternately condenses under high pressure in the condenser by surrendering heat to the environment and vaporizes under low pressure in the evaporator by absorbing heat from the medium being cooled.

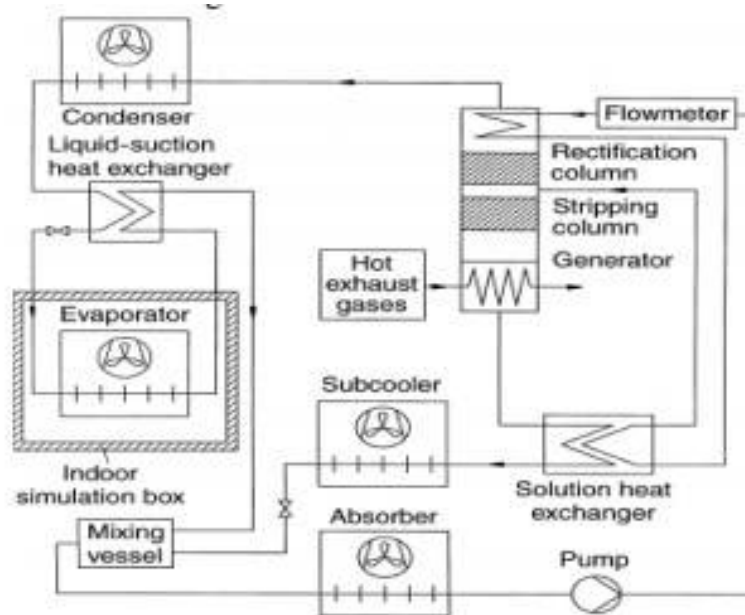


Fig. 1 VAR with exhaust gas heat recovery

The principal difference between the absorption and the vapour-compression cycles is the mechanism for circulating the refrigerant through the system and providing the necessary pressure difference between the vaporizing and condensing processes. The vapour compressor employed in the vapour-compression cycle is replaced in the absorption cycle by an absorber and a generator or boiler, which compress the vapour as required. The energy input required by the vapour-compression cycle is supplied to the compressor in the form of mechanical work but in the absorption cycle, the energy input is mostly in the form of heat supplied to the generator. In the present case the heat source is the exhaust heat of an internal combustion engine. The boiling point of ammonia is lower than that of water, so it vaporizes, separating the refrigerant from the absorbent. Since the vapour is not a pure ammonia gas, it must be purified as it flows through a stripping and rectification column. The heat exchangers of the generator rectification system were designed as compact plate-fin heat exchangers and the column was filled with stainless steel Pall rings.

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Refrigerants used for Absorption refrigeration system:

A fundamental requirement of absorbent/refrigerant combination is that, in liquid phase, they must have a margin of miscibility within the operating temperature range of the cycle. The mixture should also be chemically stable, non-toxic, and non-explosive. In addition to these requirements, the following are desirable:

- a. Refrigerant should have high heat of vaporization and high concentration within the absorbent in order to maintain low circulation rate between the generator and the absorber per unit of the cooling capacity.
- b. Transport properties that influence heat and mass transfer, e.g., viscosity, thermal conductivity, and diffusion coefficient should be favorable.
- c. Both refrigerant and absorbent should be noncorrosive, environmental friendly, and economical. There are some 40 refrigerant compounds and 200 absorbent compounds available. However, the most common working fluids are water/ammonia and LiBr/water. Since the invention of absorption refrigeration systems, water/ammonia has been widely used for both cooling and heating purposes.

The main properties are:

- a. Ammonia (refrigerant) and water (absorbent) are highly stable for a wide range of operating temperature and pressure.
- b. Ammonia has a high latent heat of vaporization, which is necessary for efficient performance of the system. Its latent heat of vaporization at -15°C is 1315kJ/Kg .
- c. Its boiling point at atmospheric pressure is -33.3°C & freezing point is -77°C .
- d. It has highest refrigerating effect per Kg of refrigerant.
- e. It is environmental friendly.

Advantages of VAR with exhaust gas recovery:

Uses Engine heat as source of energy hence enhances the efficiency of engine.

Moving parts are only in the pump, which is a small element in the system hence operation becomes smooth and also wearing and tearing is reduced.

The system works at low evaporator pressures without affecting the COP of the system.

Environmental friendly, no release of CFC derivatives.

Helps in protecting OZONE layer from depletion.

Helps engine to cool, as it extracts heat from engine. Low running cost.

Higher engine power efficiency.

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Refrigeration and Air-Conditioning (ME307)

Date:

Experiment No 11

Study of Instrumentation of all Experimental setups used in RAC Lab.

AIM: - Study of Instrumentation (Measurement Devices) of all Experimental setups used in Refrigeration and Air conditioning Lab.

INTRODUCTION: -

For getting accurate and correct results from the experiments conducted in the laboratory, it is quite important to have accurate and precise measuring instruments. Various devices used for measuring different parameters of the experimental setups are as follows:

1. Flow meter
2. Pressure gauge
3. Energy meter
4. Thermometer
5. Thermocouple

FLOW METER

Measuring the flow of liquids is a critical need in many industrial plants. In some operations, the ability to conduct accurate flow measurements is so important that it can make the difference between making a profit or taking a loss. In other cases, inaccurate flow measurements or failure to take measurements can cause serious (or even disastrous) results.

With most liquid flow measurement instruments, the flow rate is determined inferentially by measuring the liquid's velocity or the change in kinetic energy. Velocity depends on the pressure differential that is forcing the liquid through a pipe or conduit. Because the pipe's cross-sectional area is known and remains constant, the average velocity is an indication of the flow rate. The basic relationship for determining the liquid's flow rate in such cases is:

$$Q = V \times A$$

Where Q = liquid flow through the pipe

Other factors that affect liquid flow rate include the liquid's viscosity and density, and the friction of the liquid in contact with the pipe.

Direct measurements of liquid flows can be made with positive-displacement flow meters. These units divide the liquid into specific increments and move it on. The total flow is an accumulation of the measured increments, which can be counted by mechanical or electronic techniques.

The most common principals for fluid flow metering are:

- Differential Pressure Flowmeters
- Velocity Flowmeters
- Positive Displacement Flowmeters
- Mass Flowmeters

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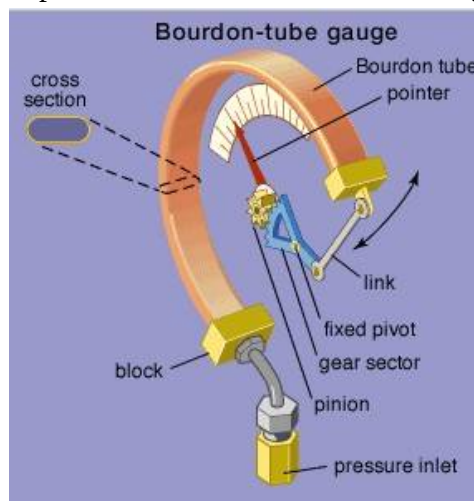
PRESSURE GAUGE

In air conditioning lab pressure gauges usually determine atmospheric pressure i.e. pressure above atmospheric. Elastic pressure sensors measure pressure using flexes, stretches, or temporarily deforms when a pressure is applied.

Elastic pressure sensors initially convert pressure into a displacement. This allows displacement sensors to be used to condition the output signal from the pressure sensor. Some pressure sensors are referred to by the method they use to measure this displacement, such as piezoelectric and capacitive pressure sensors. Where electronic displacement sensors are used, the method of detecting pressure change is usually by means of a diaphragm. Elastic pressure sensors measure differential pressure.

Bourdon tube pressure gauge:

The Bourdon tube pressure gauge is probably the most popular elastic pressure sensor. Basic Bourdon tubes are made from metal alloys such as stainless steel or brass. They consist of a tube of elliptical or oval cross-section, sealed at one end. There are various shapes of Bourdon tube, including helical, spiral and twisted. A common design is the C-shape.



Here the tube is at atmospheric pressure. When increased pressure is applied to the open end, it deflects outwards (tries to straighten) in proportion to the pressure inside the tube (the outside of the tube remains at atmospheric pressure). As the pressure is decreased, the tube starts to return to its atmospheric pressure position. The amount by which the tube moves in relation to the pressure applied to it depends on factors including its material, shape, thickness, and length. Compared to other elastic pressure sensors the deflection produced by Bourdon tubes is large. The Bourdon tube pressure gauge, shown here, consists of a Bourdon tube connected to a pointer. The pointer moves over a calibrated scale. When pressure is applied, the movement of the tube is fairly small, so to increase the movement of the pointer it is mechanically amplified. This is usually by a connecting mechanism consisting of a lever, quadrant and pinion arrangement. Bourdon tubes need some form of compensation for temperature, as temperature

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changes affect their accuracy. For remote sensing, the displacement of the Bourdon tube caused by pressure changes can be detected by a suitable displacement sensor.

ENERGY METER

An electricity meter or energy meter is a device that measures the amount of electric energy consumed by a residence, business, or an electrically powered device. Electricity meters are typically calibrated in billing units, the most common one being the kilowatt hour [kWh].

Periodic readings of electric meters establish billing cycles and energy used during a cycle. In settings when energy savings during certain periods are desired, meters may measure demand, the maximum use of power in some interval. "Time of day" metering allows electric rates to be changed during a day, to record usage during peak high-cost periods and off-peak, lower-cost, periods. Also, in some areas meters have relays for demand response shedding of loads during peak load periods.

UNITS OF MEASUREMENT

The most common unit of measurement on the electricity meter is the kilowatt hour [kWh], which is equal to the amount of energy used by a load of one kilowatt over a period of one hour, or 3,600,000 joules. Some electricity companies use the SI megajoule (MJ) instead.

Demand is normally measured in watts, but averaged over a period, most often a quarter or half hour. Reactive power is measured in "thousands of volt-ampere reactive-hours", (kVA_{rh}). By convention, a "lagging" or inductive load, such as a motor, will have positive reactive power. A "leading", or capacitive load, will have negative reactive power.

Volt-amperes measures all power passed through a distribution network, including reactive and actual. This is equal to the product of root-mean-square volts and amperes.

THERMOMETER

The mercury-in-glass or mercury thermometer was invented by physicist Daniel Gabriel Fahrenheit in Amsterdam (1714). It consists of a bulb containing mercury attached to a glass tube of narrow diameter; the volume of mercury in the tube is much less than the volume in the bulb. The volume of mercury changes slightly with temperature; the small change in volume drives the narrow mercury column a relatively long way up the tube. The space above the mercury may be filled with nitrogen or it may be at less than atmospheric pressure, a partial vacuum.

In order to calibrate the thermometer, the bulb is made to reach thermal equilibrium with a temperature standard such as an ice/water mixture, and then with another standard such as water/vapour, and the tube is divided into regular intervals between the fixed points. In principle, thermometers made of different material (e.g., coloured alcohol) might be expected to give different intermediate readings due to different expansion properties; in practice the substances used are chosen to have reasonably linear expansion characteristics as a function of true thermodynamic temperature, and so give similar results.

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In air conditioning experiment it is used to calculate dry bulb temperature and wet bulb temperature. The dry-bulb temperature (DBT) is the temperature of air measured by a thermometer freely exposed to the air but shielded from radiation and moisture. DBT is the temperature that is usually thought of as air temperature, and it is the true thermodynamic temperature. As a matter of fact, it indicates the amount of heat in the air and it is directly proportional to the mean kinetic energy of the air molecules. Temperature is usually measured in degrees Celsius ($^{\circ}\text{C}$), Kelvin (K), or Fahrenheit ($^{\circ}\text{F}$).

The wet-bulb temperature is the temperature a parcel of air would have if it were cooled to saturation (100% relative humidity) by the evaporation of water into it, with the latent heat being supplied by the parcel. An actual wet-bulb thermometer indicates a temperature close to the true (thermodynamic) wet-bulb temperature. The wet-bulb temperature is the lowest temperature that can be reached under current ambient conditions by the evaporation of water only; it is the temperature felt when the skin is wet and exposed to moving air. Wet-bulb temperature is largely determined by both actual air temperature (dry-bulb temperature) and humidity, the amount of moisture in the air.

THERMOCOUPLE

A thermocouple consists of two dissimilar conductors in contact, which produce a voltage when heated. The size of the voltage is dependent on the difference of temperature of the junction to other parts of the circuit. Thermocouples are a widely used type of temperature sensor for measurement and control and can also be used to convert a temperature gradient into electricity. Commercial thermocouples are inexpensive, interchangeable, are supplied with standard connectors, and can measure a wide range of temperatures. In contrast to most other methods of temperature measurement, thermocouples are self-powered and require no external form of excitation. The main limitation with thermocouples is accuracy; system errors of less than one degree Celsius ($^{\circ}\text{C}$) can be difficult to achieve.

Any junction of dissimilar metals will produce an electric potential related to temperature. Thermocouples for practical measurement of temperature are junctions of specific alloys which have a predictable and repeatable relationship between temperature and voltage. Different alloys are used for different temperature ranges. Properties such as resistance to corrosion may also be important when choosing a type of thermocouple. Where the measurement point is far from the measuring instrument, the intermediate connection can be made by extension wires which are less costly than the materials used to make the sensor. Thermocouples are usually standardized against a reference temperature of 0 degrees Celsius; practical instruments use electronic methods of cold-junction compensation to adjust for varying temperature at the instrument terminals. Electronic instruments can also compensate for the varying characteristics of the thermocouple, and so improve the precision and accuracy of measurements.

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Thermocouples are widely used in science and industry; applications include temperature measurement for kilns, gas turbine exhaust, diesel engines, and other industrial processes.

In 1821, the German–Estonian physicist Thomas Johann Seebeck discovered that when any conductor is subjected to a thermal gradient, it will generate a voltage. This is now known as the thermoelectric effect or Seebeck effect. Any attempt to measure this voltage necessarily involves connecting another conductor to the "hot" end. This additional conductor will then also experience the temperature gradient, and develop a voltage of its own which will oppose the original. Fortunately, the magnitude of the effect depends on the metal in use. Using a dissimilar metal to complete the circuit creates a circuit in which the two legs generate different voltages, leaving a small difference in voltage available for measurement. That difference increases with temperature, and is between 1 and 70 microvolts per degree Celsius ($\mu\text{V}/^\circ\text{C}$) for standard metal combinations.

The voltage is not generated at the junction of the two metals of the thermocouple but rather along that portion of the length of the two dissimilar metals that is subjected to a temperature gradient. Because both lengths of dissimilar metals experience the same temperature gradient, the end result is a measurement of the difference in temperature between the thermocouple junction and the reference junction.

Derivation from Seebeck effect

Upon heating, the Seebeck effect will initially drive a current. However, provided the junctions all reach a uniform internal temperature, and provided an ideal voltmeter is used, then the thermocouple will soon reach an equilibrium where no current will flow anywhere ($J=0$). As a result, the voltage gradient at any point in the circuit will be given simply by,

$$\nabla V = \mathbf{E}_{\text{emf}} = -S \nabla T$$

where S is the Seebeck coefficient at that point, and ∇T is the temperature gradient at that point.

This leads to a characteristic voltage difference independent of many details (the conductors' size, length do not matter):

$$V_{A-B} = \int_{T_c}^{T_h} (S_A(T) - S_B(T)) dT,$$

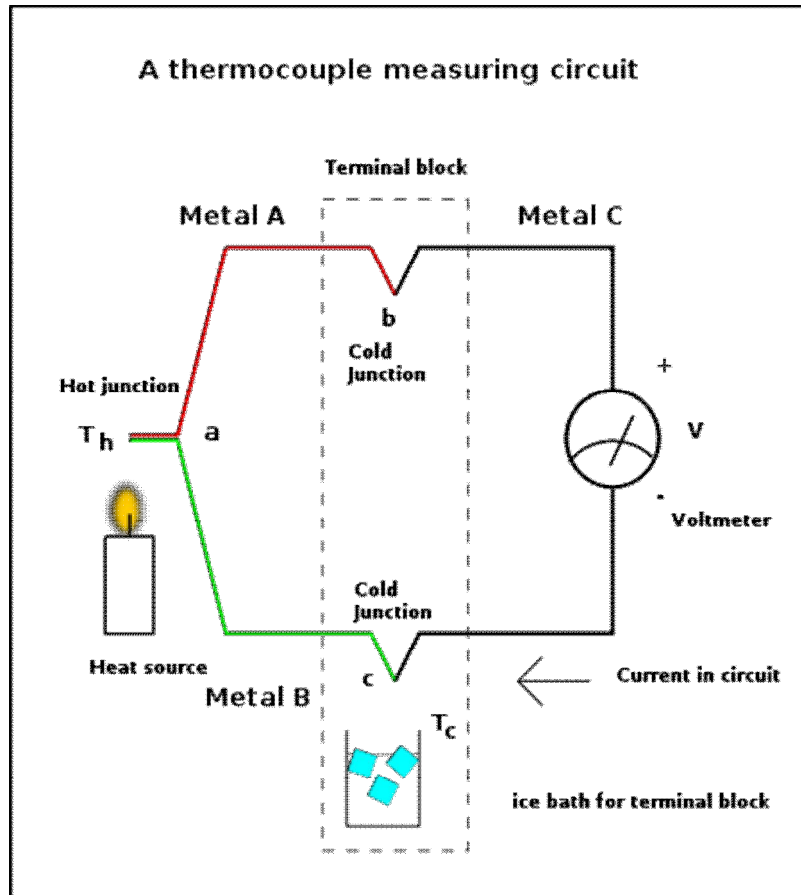
where S_A and S_B are the Seebeck coefficients of materials A and B as a function of temperature, and T_c and T_h are the temperatures of the two junctions. If the Seebeck coefficients are effectively constant for the measured temperature range, the above formula can be approximated as:

$$V_{A-B} \approx (S_A - S_B) \cdot (T_h - T_c) = \Delta S \Delta T.$$

It is important to note that the emf is not generated at the junctions themselves, but rather in the wires leading between the hot and cold junctions. As a result, the nature and composition of the

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junctions (where T is internally constant) itself does not influence the measured voltage. Conversely, if there are variations in the composition of the wires in the thermal gradient region (due to contamination, oxidation, etc.), outside the junction, this can lead to changes in the measured voltage.



UNITS:

Flow meter unit: liters per hour
Pressure gauge range: pounds per square inch
Energy meter: ampere
Thermometer: degree Celsius
Thermocouple: degree Celsius

Marks obtained:	Signature of faculty:	Date: